



Technical Service Bulletin

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New Intake Valves

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

This document introduces new intake valves for the K38, K50, QSK45 and QSK60 series engines manufactured at the Daventry Engine Plant. The new intake valves utilize a nitride hardening process known as Sursulf. The benefits of Sursulf are:

- Increased surface hardness
- Increased fatigue strength
- Improved wear resistance.

Sursulf valves can **not** be ground. Grinding will remove a significant portion of the hardened layer. Ground valves will be susceptible to 'beat in' wear and premature failure.

It is **not** recommended to intermix old intake valve part numbers with the new Sursulf valve part numbers in the same cylinder head. Cylinder head combinations are acceptable on the same engine.

The new valves have a darker appearance when compared to the old design; other markings are described below.

Note : Engines manufactured at Cummins® India Ltd. continue to use the old valve part numbers.

Intake Valve Part Numbers*		
	New	Old
K38, K50	3631950	3052820
QSK45, QSK60	3171769	4016438
* Except engines manufactured at Cummins® India Ltd.		

Note : This is a product improvement and not subject to campaign or TRP.

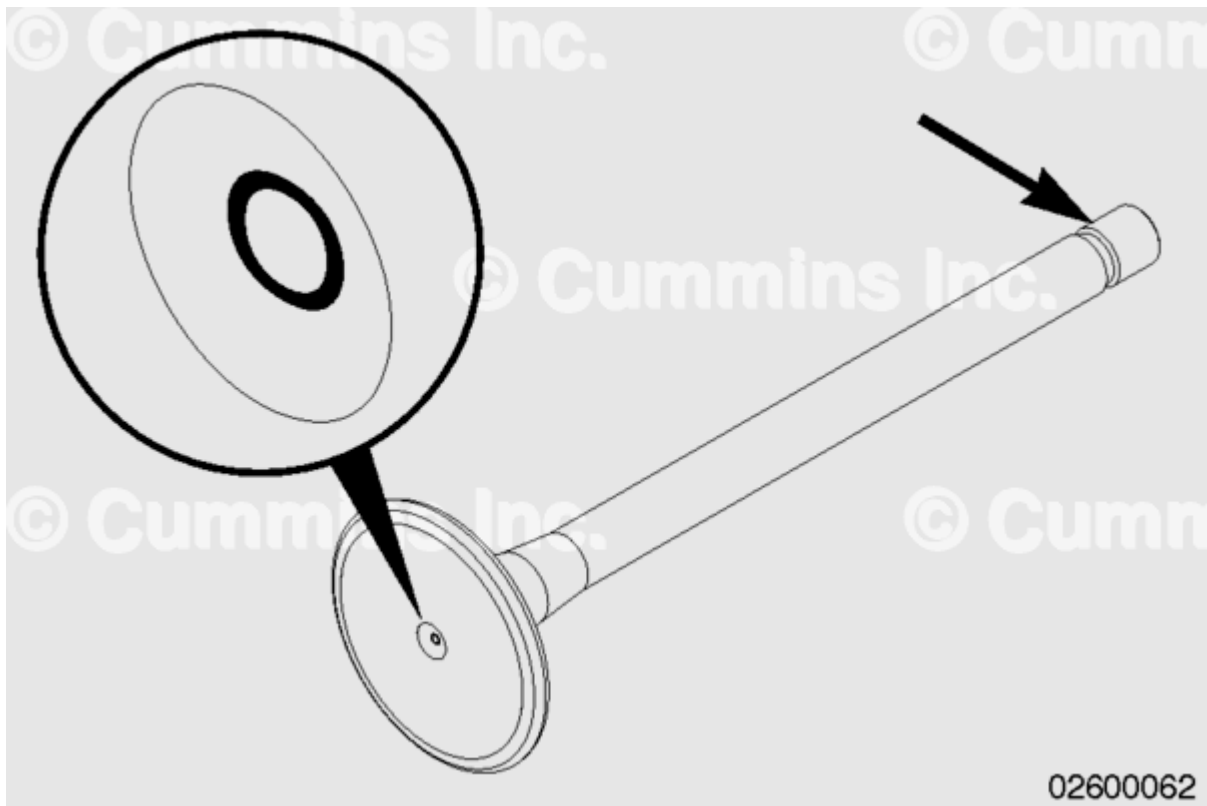


Figure 1: New K38 and K50 Intake Valve, Part Number 3631950

- The new K38 and K50 Sursulf intake valve, Part Number 3631950, is a direct replacement for intake valve Part Number 3052820.
- The new intake valve can be identified by the letter 'O' forged into the bottom, center of the valve and by the part number laser-etched into the stem, above the collet groove. See Figure 1.

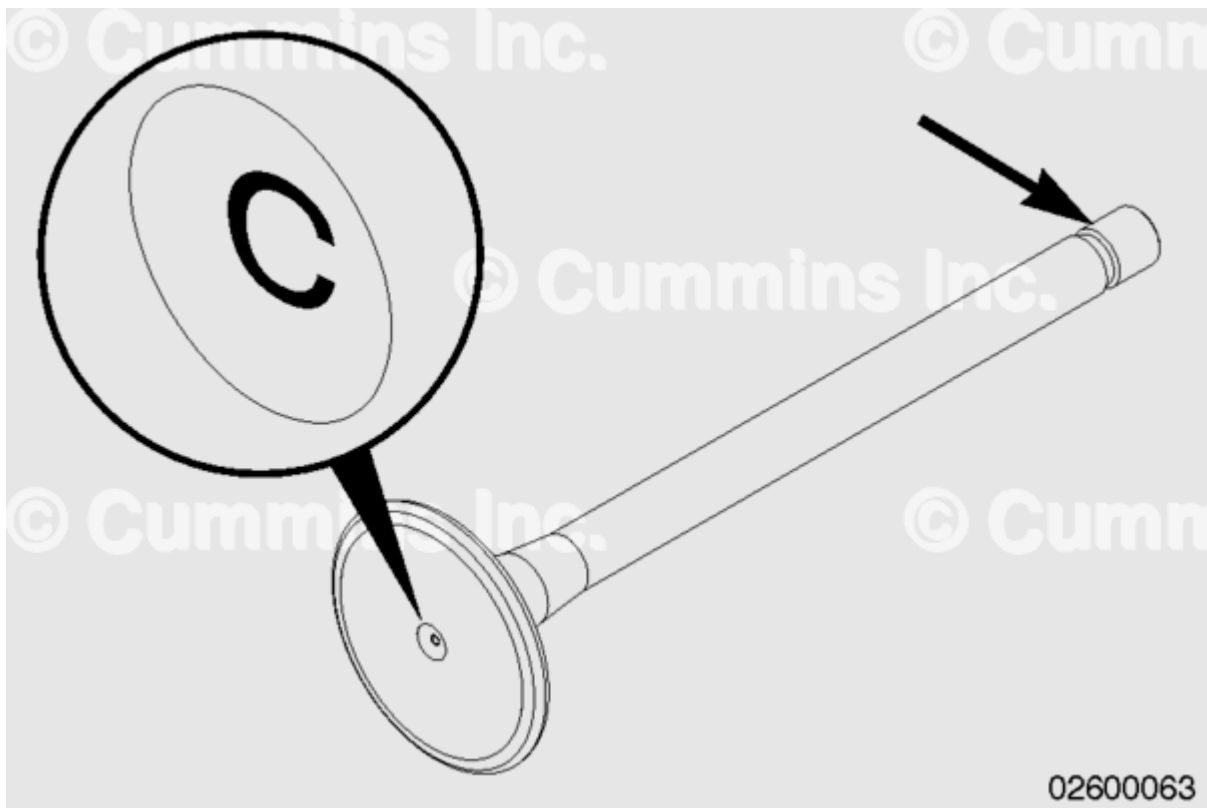


Figure 2: New QSK45 and QSK60 Intake Valve, Part Number 3171769

- The new QSK45 and QSK60 Sursulf intake valve, Part Number 3171769, is a direct replacement for intake valve Part Number 4016438.
- The new intake valve can be identified by the letter 'C' forged into the bottom, center of the valve and by the part number laser-etched into the stem, above the collet groove. See Figure 2.